

NRES 746 – Week 3, Lecture 2: A Probability Puzzle

Name: _____

Instructions: Work alone for about 5-8 minutes, guessing before you compute anything. Then compare with a neighbor.

Part 1: Sevens vs. doubles

Roll two fair six-sided dice (one red, one blue, so you can tell them apart).

1a. Before doing any counting: which do you think is more likely, rolling a **sum of 7**, or rolling **doubles** (both dice show the same number)? Circle one:

sum of 7 doubles equally likely

1b. Now check your guess. First: how many equally-likely outcomes are there in the full sample space Ω (all ordered pairs of red/blue results)?

$|\Omega| =$ _____

1c. How many of those outcomes sum to 7? List them if it helps.

$P(\text{sum} = 7) =$ _____

1d. How many outcomes are doubles?

$P(\text{doubles}) =$ _____

1e. Was your guess in 1a correct? If not, what made doubles or sevens feel more/less likely than it actually is?

Part 2: How many tickets?

A small regional lottery asks you to pick 5 distinct numbers from 1 to 30 (order doesn't matter - {3, 8, 14, 20, 29} is the same ticket as {29, 20, 14, 8, 3}). How many different tickets are possible?

Leave your answer as an expression (factorials, or $\binom{n}{k}$ notation if you already know it). No need to provide an actual integer value!

Hint: how many choices do you have for the first number you write down? How many for the second, given the first is already used? ... and then why do you have to divide by something at the end?

Part 3: A teaser (guess only – no calculation)

Suppose there are 25 people in this room. Do you think it's more likely than not (better than a 50% chance) that **at least two of them share a birthday**? Circle your guess:

yes no

We'll come back to this one once we have the tools to compute it exactly.